CYS 331 —Unit 3: Security in Data Networks

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Learning Outcomes

- Explain device-level risks that affect wireless data network security and trust.
- Describe key security ideas in Cellular Digital Packet Data and General Packet Radio Service.
- Discuss Global System for Mobile Communication protections and important architectural limitations.
- Explain how Internet Protocol Security adds stronger protection above wireless transport technologies.

Wireless Device Security Issues

- Mobile devices store credentials, data, and network trust relationships attackers can exploit.
- Lost devices, outdated software, and unsafe applications expand wireless exposure significantly.
- Weak local authentication allows unauthorized access even before network defenses activate.
- Device hardening reduces risk through updates, encryption, access control, and policy enforcement.

Cellular Digital Packet Data Security

- Cellular Digital Packet Data carried packet traffic over legacy cellular infrastructure.
- Security depended on network controls, authentication, and limited device capabilities.
- Legacy design constraints reduced flexibility for strong modern end-to-end protections.
- Historical study shows why layered security became necessary for mobile data.

General Packet Radio Service Security

- General Packet Radio Service improved packet data efficiency on mobile networks.
- It introduced subscriber authentication, temporary identifiers, and air-interface protection mechanisms.
- Security remained uneven because protection weakened beyond some network boundaries.
- Organizations still needed application-level controls above carrier-provided security.

Global System for Mobile Communication Security

- Global System for Mobile Communication used subscriber identities and challenge-response authentication.
- It protected the radio link, but did not secure every network segment.
- Unilateral authentication meant devices trusted networks more than networks trusted devices.
- Studying these limits explains later improvements in mobile security architecture.

Internet Protocol Security

- Internet Protocol Security protects network traffic with authentication, integrity, and encryption services.
- It works above link technologies, making it valuable across diverse wireless networks.
- Tunneling secures communications between gateways, while transport mode protects endpoints directly.
- Strong key management and policy design determine real Internet Protocol security effectiveness.

Practical Security Controls for Wireless Data Networks

- Combine carrier protections with endpoint hardening, virtual private networks, and monitoring.
- Protect data at rest, in transit, and during authentication processes.
- Segment sensitive traffic and apply least privilege to network access decisions.
- Plan for theft, interception, roaming, and insecure intermediate infrastructure.

Unit 3 Summary

- Wireless data security depends on endpoints, carrier architecture, and layered protections.
- Legacy mobile services offered useful protections, but left important gaps unaddressed.
- Global System for Mobile Communication reveals why mutual trust matters in network design.
- Internet Protocol Security strengthens protection across diverse wireless transport environments.